Fewer numbers, same information

- **A global SST map is 16,000 numbers a month.** An ENSO forecaster watches one: the Niño3.4 index. Most of the 16,000 move together
- **Dimensionality reduction finds the few numbers that carry the field.** This week's tool is principal component analysis, which climate science calls EOF analysis
- **Three jobs:** compress a field, discover the patterns that dominate it, and build features that a model can learn from
- **Unsupervised:** no target, no labels. The method sees only the inputs and reports how they vary

Compress
2,469 grid cells per month become 18 numbers that keep 90% of the variance

Discover
the first pattern of tropical Pacific SST is El Niño, found without being told where to look

Prepare
10 principal components forecast Niño3.4 six months ahead better than persistence. 2,469 raw cells cannot

Following the Week 7 lecture notes on dimensionality reduction and the tutorial on tropical Pacific SST

The curse of dimensionality, on real data

- **The task:** predict Niño3.4 six months ahead from this month's tropical Pacific SST anomalies. Train on 1950 to 2000, test on 2001 to 2024
- **Linear regression on all 2,469 cells:** training error 0.32 °C, test error 1.49. Worse than predicting zero. With more features than months, a line fits every training point and learns nothing
- **Linear regression on 10 principal components:** test error 0.74, better than persistence and than climatology
- **The same model, the same data.** The only change is the number of inputs, and the lesson is that a model needs samples per feature, not features per sample

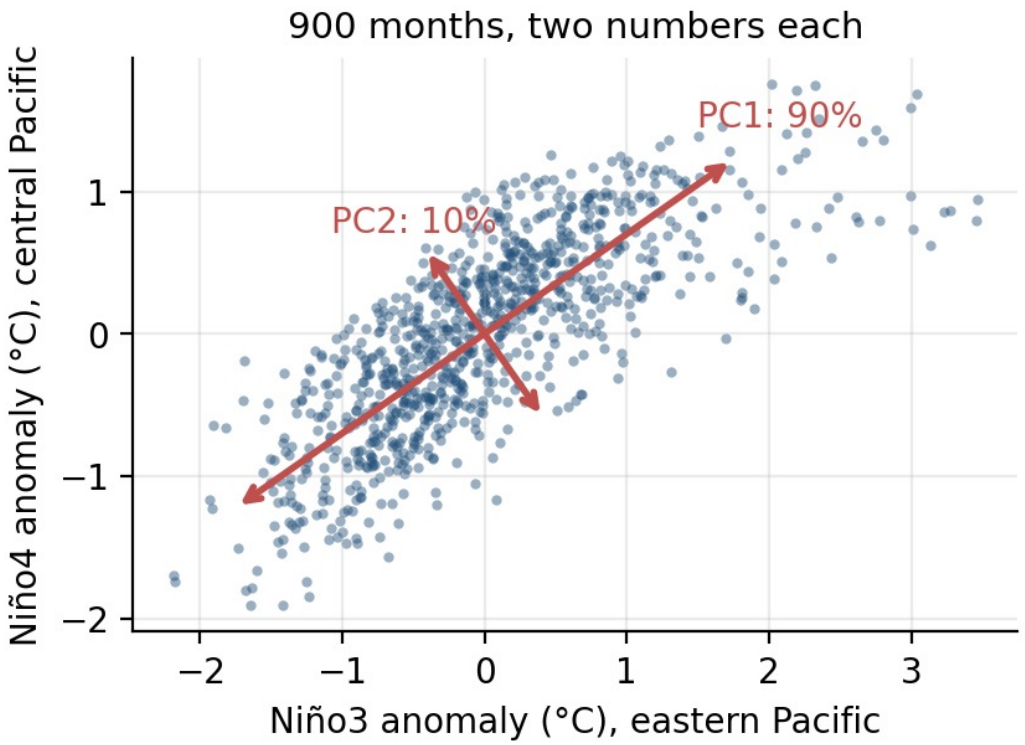
Six months ahead, RMSE (°C)	Train	Test
Climatology, predict zero		0.85
Persistence, this month's value		0.96
Regression on 2,469 grid cells	0.32	1.49
Ridge on 10 principal components		0.74

Why high dimensions hurt. With 612 training months and 2,469 inputs there are infinitely many exact fits, and the one the solver picks memorizes the noise. Distances between samples also stop meaning much: in 2,469 dimensions every month is about equally far from every other, so the nearest neighbor is barely nearer.

The fix is not more data. Seventy-five years is all the record there is. The fix is fewer, better inputs.

NOAA ERSST v5 monthly SST, tropical Pacific 30S to 30N, 120E to 60W, 1950 to 2024, 900 months. Test years 2001 to 2024, lead of six months. Persistence is this month's Niño3.4 anomaly carried forward

What PCA does: the geometry



- **Two numbers per month:** the SST anomaly in the eastern Pacific (Niño3) and in the central Pacific (Niño4). Each month is a dot
- **They are redundant.** When the east is warm the center is warm. The cloud is a tilted ellipse, not a circle
- **PCA finds the long axis of the ellipse.** The direction of largest variance is PC1, and it carries 90 percent of it. The direction at right angles is PC2, with the rest
- **Project onto PC1 and you keep one number per month** instead of two, and lose a tenth of the variance
- **The field version is the same picture** with 2,469 axes instead of two. Nobody can draw it, but the arithmetic does not care

NOAA ERSST v5 monthly SST, tropical Pacific 30S to 30N, 120E to 60W, 1950 to 2024, 900 months. Niño3: 5S to 5N, 150W to 90W. Niño4: 160E to 150W. Arrows are two standard deviations long

The recipe, and the two vocabularies

- **Arrange the data as a matrix X:** one row per sample, one column per feature. For a field, one row per month and one column per grid cell
- **Center:** subtract each column's mean. PCA describes variation around the mean, not the mean itself
- **Factor:** the singular value decomposition $X = U S V^T$. The rows of V^T are the components, unit vectors in feature space, ordered by the variance they capture
- **Project:** X V gives the scores, the coordinates of each sample along each component
- **Truncate:** keep the first k components. Rebuilding X from them gives the best rank-k approximation of the data, in the least-squares sense

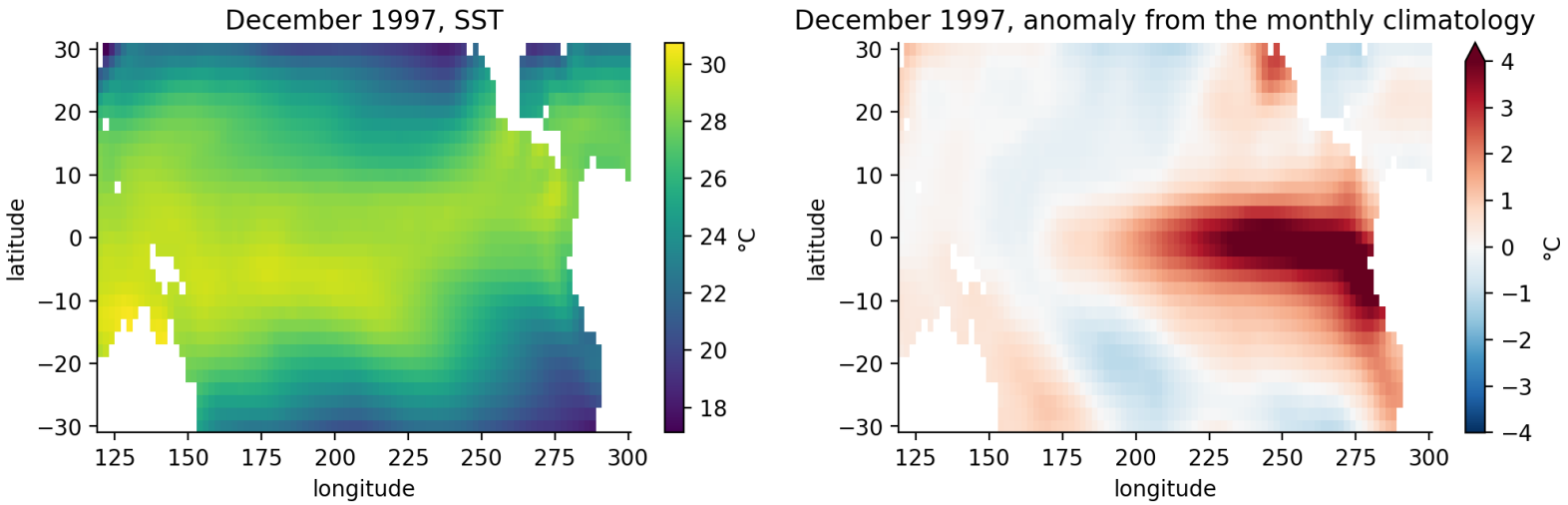
```
pca = PCA(n_components=10).fit(X)      # X: months by cells, centered
scores = pca.transform(X)              # months by 10: the PCs
patterns = pca.components_             # 10 by cells: the EOFs
share = pca.explained_variance_ratio_  # fraction of variance each
```

Statistics and ML	Climate science	In scikit-learn
principal component (direction)	EOF, the spatial pattern	components_[k]
score (coordinate)	principal component, PC, the time series	transform(X)[:, k]
explained variance	eigenvalue of the covariance	explained_variance_
share of variance	fraction of variance explained	explained_variance_ratio_
loading	loading, weight	entry of components_

One calculation, two names. PCA was invented by statisticians in 1901. Meteorologists reinvented it in 1956 as EOF analysis and kept their own words. The confusing one is "principal component": in ML it is the direction, in climate it is the time series. Context usually settles it.

Pearson 1901, Hotelling 1933; Lorenz 1956 for the EOF name. Lecture notes, section on PCA, for the SVD

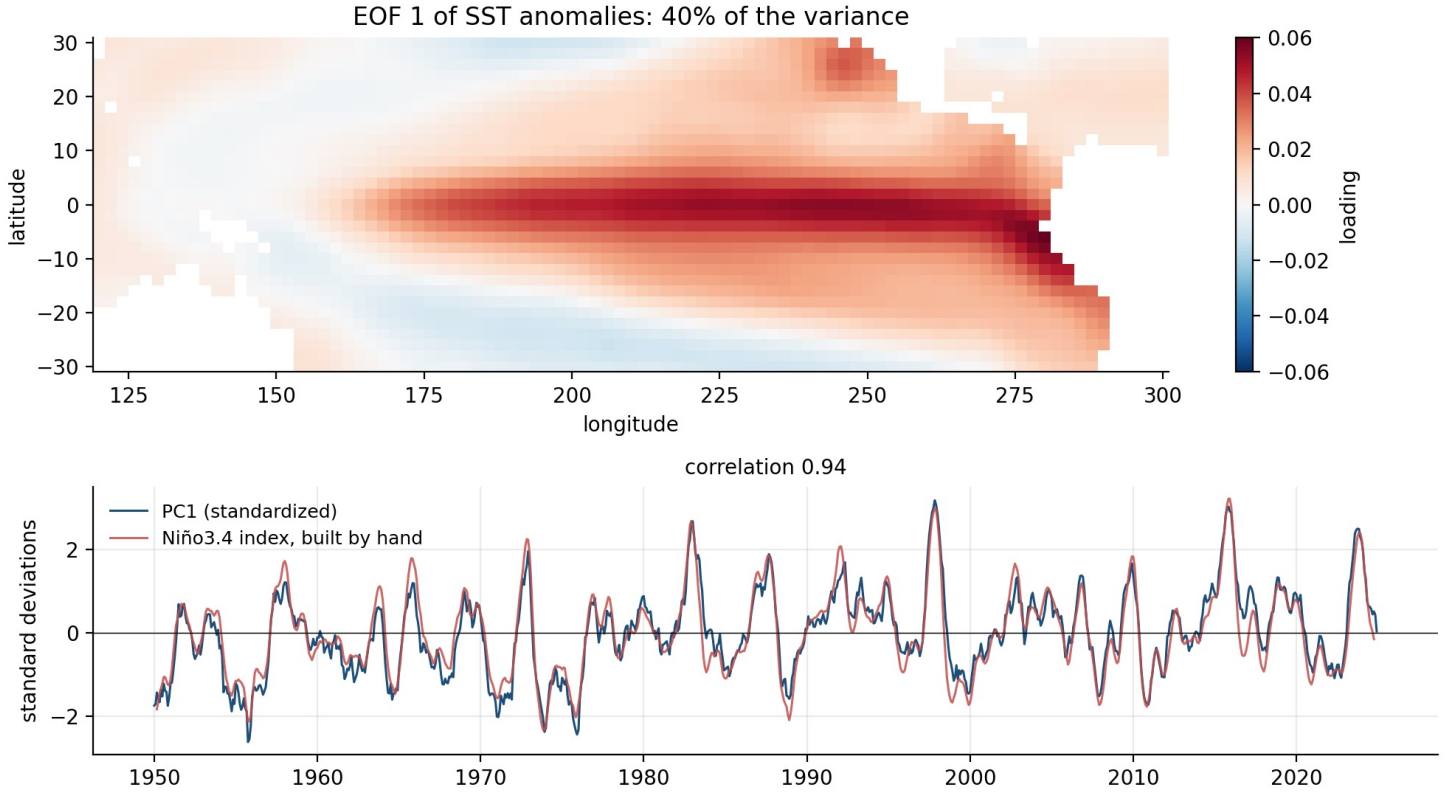
Why sea surface temperature



- **The ocean is the climate system's memory.** Water holds heat for months, so SST anomalies persist and the atmosphere above them responds. Seasonal forecasts are built on SST
- **ENSO is the largest year-to-year signal on Earth.** The central and eastern Pacific warm by a degree or more for a year, and rainfall shifts on every continent
- **A field, not a point.** The raw map says little: the tropics are warm everywhere. The anomaly, this month minus a normal month, shows the event. December 1997, the strongest El Niño on record, is 3 to 4 degrees above normal along the equator
- **The question for PCA:** given 900 such maps and no hint where to look, does it find this pattern?

NOAA ERSST v5 monthly SST, tropical Pacific 30S to 30N, 120E to 60W, 1950 to 2024, 900 months. Left: SST. Right: anomaly from the 1950 to 2024 climatology for December. Land is white

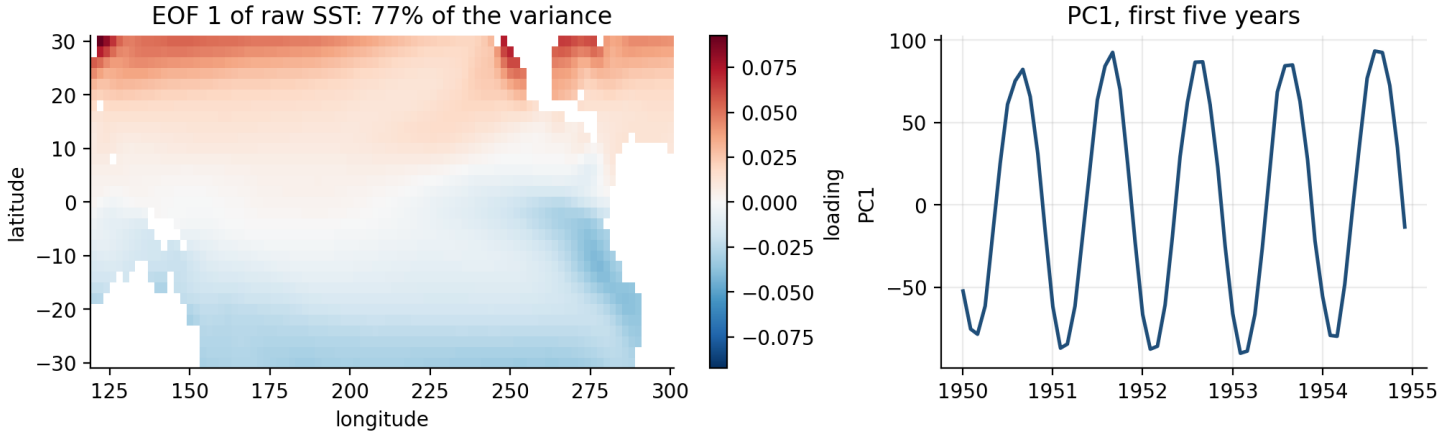
PCA finds El Niño on its own



- **EOF 1 carries 40 percent of the variance** of 2,469 grid cells. EOF 2 has 15, and then a long tail
- **The map is El Niño:** warm along the equator from the dateline to South America, strongest in the east, with a faint horseshoe of cooling to the west
- **Its time series matches the Niño3.4 index at 0.94.** The index was built by hand from a box a scientist drew in the 1990s. PCA drew nothing. It found the pattern because the pattern is where the variance is
- **Every big event is there:** 1972, 1982 to 83, 1997 to 98, 2015 to 16, 2023 to 24. The index is smoother only because of its 5-month running mean
- **This is discovery, not prediction.** PCA was told nothing about ENSO. It found the dominant mode of a field, which is the job climate science has used it for since the 1950s

NOAA ERSST v5 monthly SST, tropical Pacific 30S to 30N, 120E to 60W, 1950 to 2024, 900 months. Area weighted by the square root of cosine latitude, land cells dropped. The Niño3.4 index is the anomaly over 5S to 5N, 170W to 120W, 5-month running mean, standardized

Three habits for fields, and one thing not to do



On the raw field the first pattern is the seasonal cycle, 77 percent of the variance, with a clean annual wave for a time series. True, and useless. PCA ranks whatever varies most, so remove what you already understand before asking what is left.

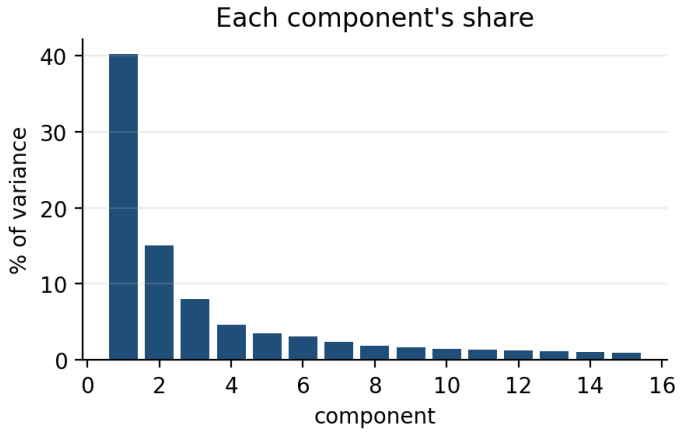
The reshape is where the work is. A (time, lat, lon) array becomes (time, lat × lon) with reshape, the land columns are removed with a boolean mask, and every EOF comes back as a vector that the same mask turns into a map. The tutorial's to_matrix and to_map do exactly this, and you will reuse them.

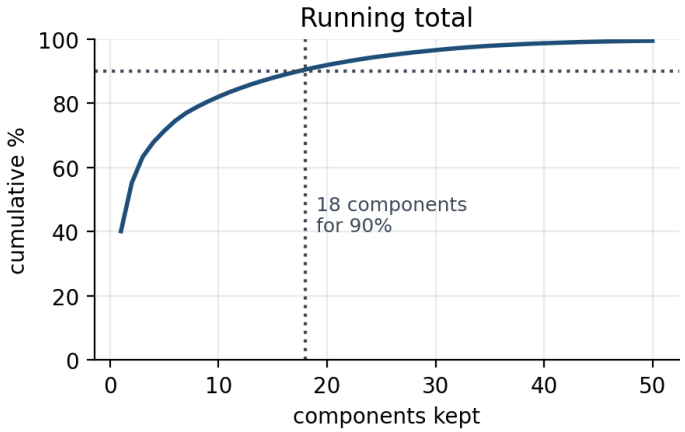
Standardizing is right for a table of different variables, temperature next to pressure next to humidity, where the units are incomparable. It is wrong for one field, where the variance differences between cells are the signal.

- **Anomalies first.** Subtract the climatology for each calendar month. If a trend is not the question, remove that too, or it will take a component of its own
- **Weight by area.** A grid cell at 60N covers half the area of one at the equator. Multiply each cell by the square root of cosine latitude so that variance, which is squared, is weighted by area
- **Drop the land, do not fill it.** Keep a mask of ocean cells, run PCA on those columns, and use the mask to put the pattern back on the map. Filling with zero adds hundreds of fake features that never vary
- **Do not standardize the cells.** StandardScaler would give a quiet cell in the western Pacific the same weight as a cell in the cold tongue that swings by four degrees, and erase the thing you are looking for. For one field in one unit: center, do not scale

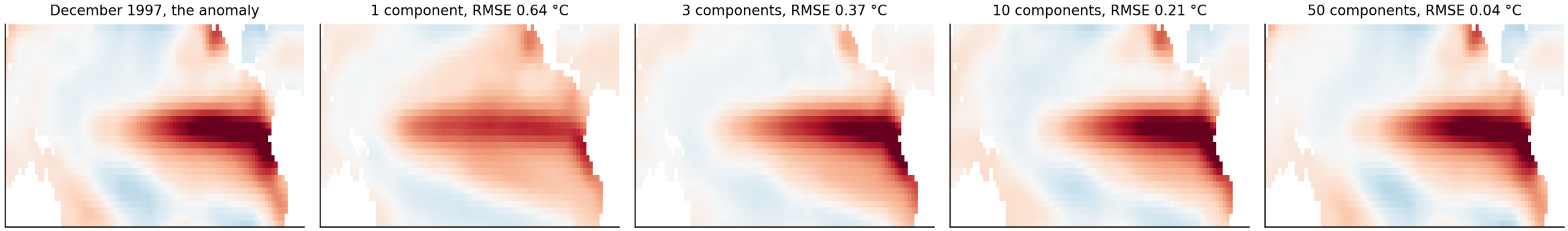
NOAA ERSST v5 monthly SST, tropical Pacific 30S to 30N, 120E to 60W, 1950 to 2024, 900 months, before removing the seasonal cycle. PC1 shown for 1950 to 1954

How many components to keep





- **The spectrum is the decision aid.** A steep drop and a long tail. The first component has 40 percent, the first ten 82, and 90 percent takes 18
- **PCA(n_components=0.9)** keeps as many as needed for 90 percent. The threshold is a choice, not a law
- **Look at what you lose.** December 1997 from 1 component has the shape, RMSE 0.64 °C. Three fix the east, 0.37. Ten are close, 0.21. Fifty are the original, 0.04
- **2,469 numbers to 10,** with a fifth of a degree of error. That ratio is why PCA is the first compression of any climate field



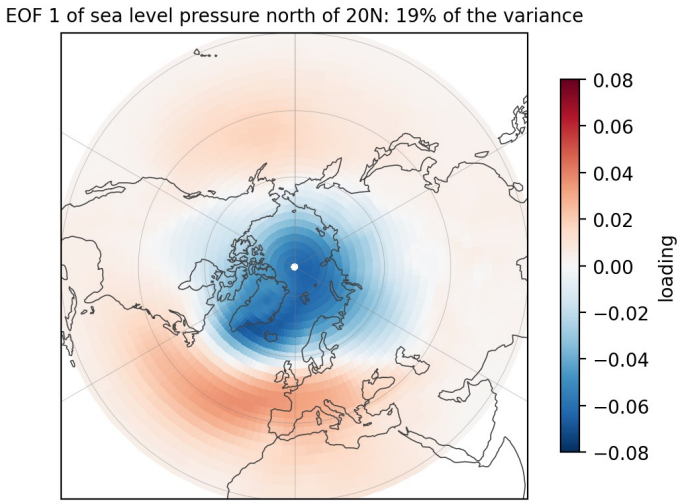
NOAA ERSST v5 monthly SST, tropical Pacific 30S to 30N, 120E to 60W, 1950 to 2024, 900 months. Bottom row: the December 1997 anomaly and its reconstruction from the first k components, same color scale. RMSE over ocean cells

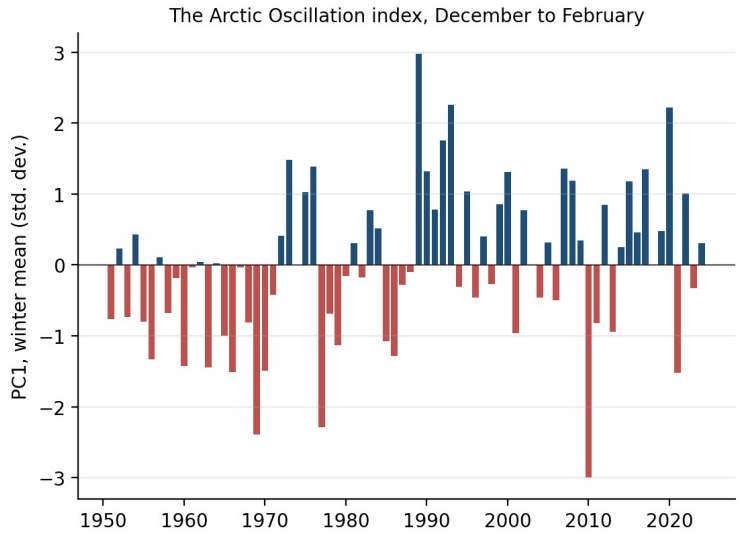
Reading the patterns: what an EOF is and is not

- **The sign is arbitrary.** A pattern and its negative describe the same variance. Fix the sign by a convention, warm Niño3.4 means positive, and say which you chose
- **Orthogonal is imposed, not discovered.** Each EOF is at right angles to the ones before it because the method demands it. The ocean is under no such obligation, so EOF 2 is not automatically a second physical mode
- **Close eigenvalues mean mixed patterns.** When two components explain nearly the same variance, PCA cannot tell them apart, and each pattern is a mixture of the two. North's rule of thumb: the separation should exceed the sampling error of the eigenvalue
- **Variance is not importance.** The first component is the biggest, not the most interesting. A small component can be the one that matters for your question
- **Linear only.** An EOF is a fixed pattern that grows and shrinks. A signal that moves, a wave propagating along the equator, needs two EOFs in quadrature to describe it, and neither looks like the wave
- **The domain is a choice.** Change the box and the patterns change. An EOF is a property of the data you gave it, not of the planet

North, Bell, Cahalan and Moeng 1982 for the eigenvalue rule. Hannachi, Jolliffe and Stephenson 2007 for a review of EOFs in atmospheric science

The same trick, across the atmosphere and ocean





- **Many of the named modes of climate variability are an EOF.** The index is PC1, and the map is how the mode is defined
- **Arctic Oscillation:** EOF 1 of sea level pressure north of 20N, shown here, 19 percent of the variance. Low over the pole, high at mid-latitudes. Its sign sets the winter over Europe and the eastern United States
- **North Atlantic Oscillation:** the same pattern cut to the Atlantic sector
- **Southern Annular Mode:** the Antarctic twin, EOF 1 of Southern Hemisphere pressure or geopotential height
- **Pacific Decadal Oscillation:** EOF 1 of North Pacific SST, defined in 1997, in this week's assignment
- **Madden-Julian Oscillation:** the two leading EOFs of tropical cloud and wind combined. The pair describes a pattern that moves east, the propagating case from the last slide

NCEP-NCAR Reanalysis 1 monthly sea level pressure, 20N to 87.5N, 1950 to 2024, monthly anomalies, area weighted. Right: winter mean of PC1, December to February

PCA as a step inside a pipeline

Features for a forecast
Ten PCs of Pacific SST forecast Niño3.4 six months ahead at 0.74 °C, against persistence 0.96. Statistical seasonal forecasts and downscaling schemes use PCs of large-scale fields as their inputs

Emulating a climate model
Compress the output field to a few dozen PCs, fit the emulator to those, rebuild the map at the end. Week 13 does this with ClimateBench. Pattern scaling is the one-component version

Weather forecasting and data assimilation
Ensemble forecasts and assimilation schemes work in reduced spaces: the leading directions of the error covariance, which are EOFs of the ensemble

Paleoclimate reconstruction
Project a handful of proxy records, tree rings, corals, ice cores, onto the EOFs of the modern field to rebuild past maps from sparse points

Air quality source apportionment
PCA and its cousin positive matrix factorization on aerosol composition: the components are sources, traffic, dust, sea salt, wood smoke, and the scores are their daily contributions

Remote sensing
Hyperspectral and multi-channel satellite data: hundreds of channels compressed to a few before classification. Cloud regimes from histograms of cloud properties

- **The common shape:** unsupervised first, then supervised. PCA turns a field into a few uncorrelated numbers, and the model that follows gets inputs it can learn from
- **The Week 6 rules still apply.** Fit the PCA on the training years and transform the test years with it. Put it in the Pipeline so cross-validation refits it inside each fold. PCA that has seen the test set is leakage, the same as a scaler
- **A neural network can learn the compression too,** an autoencoder, Week 12. PCA is the linear autoencoder, and it is the honest first attempt

Forecast numbers from the Week 7 tutorial, section 9. Pipeline leakage: Week 6, slide on leakage through preprocessing

When PCA is the wrong tool, and what comes next

- **The structure is nonlinear.** A curve, a ring, a folded sheet. PCA fits a flat plane and smears it. Kernel PCA, manifold methods, and autoencoders follow curves
- **The signal moves.** Propagating waves need extended or complex EOFs, or a method that models time, which is where the RNN of Week 11 came in
- **The physical modes are not orthogonal.** Rotated EOFs, varimax, trade orthogonality for patterns that are localized and easier to read, at the cost of the variance ordering
- **Sources must be non-negative.** Mass from a pollution source cannot be negative. Non-negative matrix factorization and PMF add that constraint and the components become physical
- **You want to see the data, not model them.** t-SNE and UMAP make two-dimensional pictures that preserve neighborhoods, good for looking, not for arithmetic. Week 9 uses them with clustering
- **Variance is not what you care about.** If the target is known, a supervised reduction, partial least squares, finds directions that predict it. If the goal is independence rather than variance, ICA

Before you reach for an alternative

Have you removed the seasonal cycle and the trend? Weighted by area? Checked that the leading eigenvalues are separated? Looked at the reconstruction? Most bad EOFs are a preparation problem, not a method problem.

This week

Tutorial: PCA on tropical Pacific SST, the El Niño pattern, and the forecast from PCs. Assignment: the Pacific Decadal Oscillation, and one preparation step that changes everything. Next week: clustering, the other half of unsupervised learning.

Lecture notes, sections on when PCA does not work well and where next. Hannachi et al. 2007 for rotated and extended EOFs

Extra

The linear algebra behind PCA

- **Center the n by p data matrix X.** The covariance is $C = X^T X / (n - 1)$, p by p, symmetric
- **Eigen-decompose:** $C\,v = \lambda\,v$. The eigenvectors v are the components, the eigenvalues λ the variance along each. Symmetric means the eigenvectors are orthogonal and the eigenvalues real
- **Or skip the covariance:** the SVD $X = U\,S\,V^T$ gives the same V, and $\lambda = s^2 / (n - 1)$. Never forming C matters when p is 2,469 or a million
- **Scores:** $T = X\,V = U\,S$. Column k of T is the k-th PC time series
- **Rank-k reconstruction:** $\hat{X} = T[:, :k]\,V[:, :k]^T$, the best rank-k approximation in squared error (Eckart and Young, 1936)
- **Fraction of variance:** $\lambda_k / \Sigma\,\lambda$. For a field with $n < p$ there are at most $n - 1$ nonzero eigenvalues, which is why 900 months give at most 899 EOFs of 2,469 cells

```
Xc = X - X.mean(axis=0)
U, s, Vt = np.linalg.svd(Xc,
        full_matrices=False)
eofs = Vt            # rows: patterns
pcs = U * s          # columns: PCs
var = s**2 / (len(X) - 1)
share = var / var.sum()
```

scikit-learn's PCA is this SVD with the centering done for you, a randomized solver for large matrices, and the attributes on the recipe slide. The results are identical up to the sign of each component.

Area weights multiply the columns of X before the SVD. The EOF comes back in weighted units, so divide by the weight to plot it.

Lecture notes, section on PCA. Jolliffe, Principal Component Analysis, for everything else

PCA in scikit-learn, in one place

```
from sklearn.decomposition import PCA

pca = PCA(n_components=10)          # or 0.9 for "enough for 90%"
pca.fit(X_train)                     # centers, then SVD; train rows
T_train = pca.transform(X_train)     # scores, n_train by 10
T_test  = pca.transform(X_test)      # same patterns applied to new rows

pca.components_                      # 10 by p: the EOFs
pca.explained_variance_ratio_        # share of variance, sums to <= 1
pca.n_components_                    # how many were kept
pca.mean_                            # the column means it subtracted
X_hat = pca.inverse_transform(T_test)   # rebuild from 10

# inside a supervised pipeline, refit in every fold
model = make_pipeline(PCA(n_components=10), Ridge(alpha=1.0))
cross_val_score(model, X, y, cv=TimeSeriesSplit(5))
```

- **fit on training rows only,** then transform both. The components are a fitted object, like a scaler's mean
- **n_components** takes an integer, a fraction between 0 and 1 for a variance target, or "mle" for an automatic choice that rarely agrees with anyone
- **svd_solver="randomized"** is picked automatically for large matrices and is what makes a million-cell field feasible. IncrementalPCA fits in batches when X does not fit in memory
- **Trailing underscores** mark what fit learned. No underscore, something you chose
- **The sign of components_[k]** can differ between scikit-learn versions and solvers. Fix it yourself, every time, before comparing to a published index

scikit-learn 1.7, sklearn.decomposition.PCA. The pipeline line is the Week 6 leakage rule applied to PCA